CS 578 Final CAFNT Project

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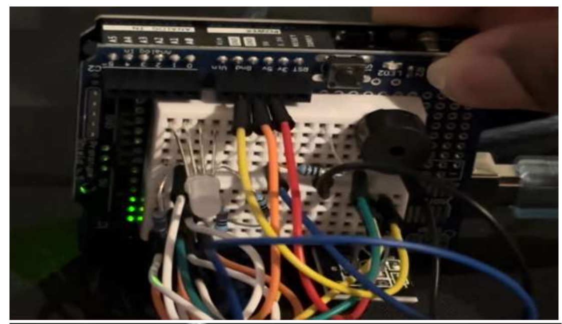
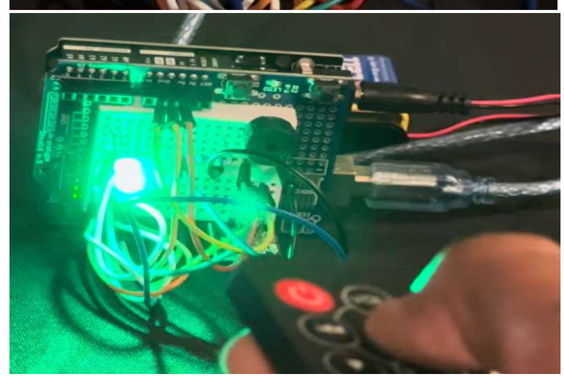
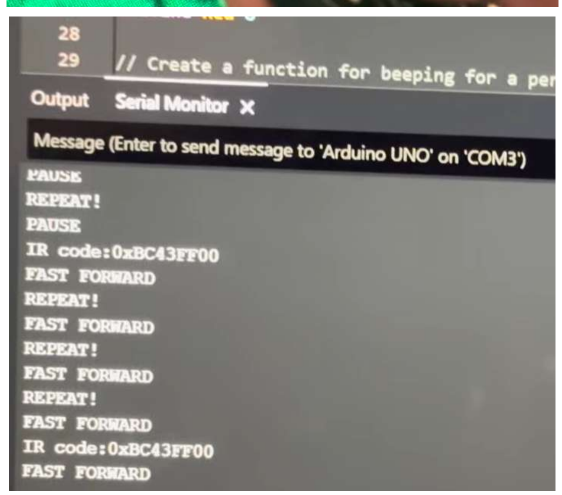
Project overview:

The CAFNT (Common API for Network Tunneling) project demonstrates a prototype system for transmitting and interpreting commands using infrared (IR) communication and serial interfaces. The system is designed to simulate long-distance communication by using an IR remote as a transmitter and an Arduino-based receiver as the processing unit.

The goal of this project is to develop a unified API that allows commands received from different input sources (IR signals or serial communication) to control hardware behavior in a consistent and modular way.







System Components (prototype):

- IR Remote (Transmitter): Used to simulate external command input
- Arduino (Receiver): Processes incoming IR signals
- Breadboard Circuit: Includes RGB LED and buzzer for output feedback
- Serial Monitor / Daemon: Displays incoming data and allows debugging

Source Code (GitHub:):

- Arduino Code (IR_Receiver_Blueprint.ino): translateIR(), pushSwitch(), Controls hardware
- Serial Daemon (Serial Daemon.cpp): Accepts user commands from the PC, receives data via serial communication

Conclusion:

As a result, our system demonstrates how a remote control can act as a transmitter while an Arduino processes incoming signals and controls hardware components such as LEDs and a buzzer. One key outcome is the creation of a simple API that connects different input methods to the same set of functions. By translating IR signals into mode codes and mapping them to Arduino functions, both IR input and serial commands can control the system consistently.